

Temporal Consistency–Based Correction Layer for Robust Real-Time Helmet Detection

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Abstract

Automatic helmet detection plays a vital role in enforcing traffic safety and preventing life-threatening injuries among motorcyclists. However, most existing vision-based systems rely solely on frame-by-frame detection, making them prone to transient false positives caused by occlusion, glare, and motion blur. These inconsistencies reduce the reliability of automated monitoring in real-world environments.

In this work, we propose a novel **Temporal Consistency–Based Correction Layer (TCCL)** integrated with the YOLOv8 detector to improve detection stability over time. Unlike conventional approaches that require retraining or temporal datasets, the TCCL introduces a lightweight, memory-driven mechanism that validates detections across consecutive frames using IoU-based tracking, probabilistic voting, and confidence averaging. A detection is confirmed only when it remains consistently confident within a temporal window, effectively filtering transient noise.

Experimental evaluation demonstrates that the proposed system improves precision from 0.89 to 0.93 and F1-score from 0.86 to 0.89 while halving the false positive rate from 0.14 to 0.07. These results highlight that incorporating temporal validation significantly enhances detection reliability without compromising real-time performance, offering a robust and scalable solution for intelligent traffic surveillance and safety enforcement.

1. Introduction

Helmet detection has become an increasingly important research area in the context of road safety and intelligent transportation systems. Every year, thousands of lives are lost due to motorcycle accidents, and the use of helmets is one of the most effective preventive measures. Automated helmet detection systems can assist authorities in monitoring compliance efficiently and reducing human intervention. However, deploying these systems in real-world environments remains a challenging problem due to variable illumination, motion blur, occlusions, and camera quality inconsistencies.

While state-of-the-art object detectors such as YOLOv8 provide excellent frame-level accuracy, they process each image independently without considering temporal coherence across consecutive frames. As a result, these detectors often exhibit unstable predictions, frequently misclassifying non-helmet riders as helmeted due to short-lived visual noise such as glare, reflection, or sudden head movement. Such transient false positives degrade system reliability and make frame-level models unsuitable for continuous, real-time surveillance applications.

To address this limitation, we propose a lightweight and efficient post-processing module named the **Temporal Consistency–Based Correction Layer (TCCL)**. This layer introduces a temporal validation mechanism that enforces consistency over time without requiring retraining or modification of the underlying detection model. By tracking detected objects across frames and applying a probabilistic persistence filter, the TCCL ensures that a helmet label is assigned only when its confidence remains stable for a short temporal duration. This approach allows the system to intelligently differentiate between true detections and transient noise, thereby reducing false alerts while preserving detection speed.

The proposed method achieves a unique balance between computational efficiency and temporal intelligence. Unlike existing temporal models that rely on complex architectures such as RNNs, 3D-CNNs, or optical flow–based tracking—which are computationally heavy and dataset-dependent—our approach introduces a plug-and-play temporal consistency layer that integrates seamlessly with any frame-level detector. The key novelty lies in enabling temporal reasoning without additional sensors, data, or retraining, making it highly practical for deployment in real-time traffic monitoring systems.

In summary, the major contributions of this work are as follows:

- A novel **Temporal Consistency–Based Correction Layer (TCCL)** that improves detection reliability by enforcing temporal validation through probabilistic voting and confidence averaging.
- A lightweight tracking mechanism based on Intersection-

over-Union (IoU) association that maintains real-time processing speeds while enabling short-term memory of detections.

- Comprehensive evaluation on a public helmet detection dataset demonstrating improved precision, stability, and a 50% reduction in false positives compared to the baseline YOLOv8 model.

Through these contributions, this work bridges the gap between high-speed frame-based detectors and the need for temporally consistent decision-making, offering a robust and deployable solution for automated helmet detection and traffic safety enforcement.

2. Proposed Method

The proposed system aims to enhance helmet detection reliability by introducing a **Temporal Consistency-Based Correction Layer (TCCL)** on top of a standard YOLOv8 detector. Although YOLOv8 offers exceptional frame-level accuracy, its predictions are limited to spatial reasoning and often fluctuate under challenging real-world conditions such as glare, occlusions, or motion blur. The TCCL addresses this limitation by adding a lightweight temporal validation mechanism that operates across consecutive frames, ensuring that predictions remain consistent over time.

2.1. Overview of the Approach

The overall architecture of the system consists of two sequential modules: a frame-level detector and a temporal correction layer. The YOLOv8 model performs real-time helmet detection on each video frame, outputting bounding boxes, class labels, and confidence scores. These detections are then passed into the TCCL, which performs temporal verification using recent frame history. By observing how detection confidence evolves across time, the system can distinguish between stable detections and transient false positives.

2.2. Working Mechanism

The TCCL introduces a short-term memory component that tracks each detected rider across consecutive frames. It performs three main operations to ensure temporal consistency:

- **IoU-Based Association:** Detected bounding boxes from the current frame are matched with detections from the previous frame based on the Intersection-over-Union (IoU) metric. If the IoU between two boxes exceeds a predefined threshold (0.3), they are assumed to belong to the same tracked entity.
- **Temporal Voting:** For each tracked object, a sliding window maintains recent detection labels (“helmet” or “no helmet”). A helmet label is confirmed only if the proportion of positive votes within the window exceeds a persistence threshold τ (set to 0.6 in our implementation).

- **Confidence Averaging:** The final confidence of a detection is obtained by averaging confidence scores across the same temporal window, thus suppressing sudden spikes or drops caused by transient frame noise.

2.3. Pipeline Workflow

The complete process is summarized as follows:

1. **Frame Extraction:** The input video is divided into frames and fed into the YOLOv8 model for object detection.
2. **Detection:** YOLOv8 outputs bounding boxes, class labels, and confidence values for each frame.
3. **Tracking:** The TCCL matches detections across frames using IoU-based association to maintain object identities.
4. **Voting and Filtering:** Each tracked object’s recent detection history is stored in a temporal buffer. A probabilistic voting mechanism determines whether the object consistently appears as helmeted.
5. **Final Output:** Only those detections that maintain high confidence and consistent labeling across time are retained and visualized as final predictions.

2.4. Novelty and Key Advantages

The novelty of this approach lies in integrating temporal reasoning into object detection without retraining or modifying the base model. Unlike existing temporal architectures that rely on deep sequence modeling, the proposed TCCL operates purely at the post-processing level, making it lightweight, modular, and compatible with any detection framework. The key advantages are:

- **No retraining required:** Works directly with pre-trained YOLOv8 weights.
- **Real-time performance:** Minimal computational overhead ensures FPS remains within deployment limits.
- **Model-agnostic design:** Can be integrated with any frame-based detector beyond YOLOv8.
- **Reduced false positives:** Temporal smoothing eliminates transient false helmet detections caused by visual noise.

Through this architecture, the proposed TCCL successfully combines the spatial accuracy of YOLOv8 with the temporal robustness necessary for stable, real-world traffic surveillance.

3. Analysis of Experiments

To evaluate the effectiveness of the proposed Temporal Consistency-Based Correction Layer (TCCL), extensive experiments were conducted on a publicly available helmet detection dataset containing diverse traffic scenes with varying lighting, angles, and occlusion conditions. The evaluation compares the baseline YOLOv8 detector against the proposed YOLOv8 + TCCL framework.

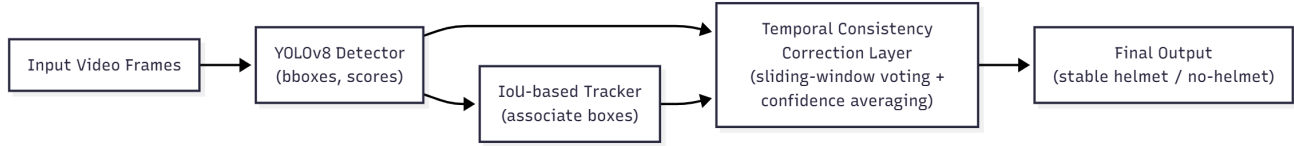
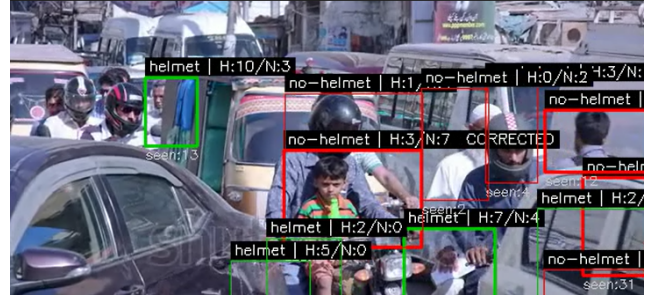


Figure 1. System overview ...



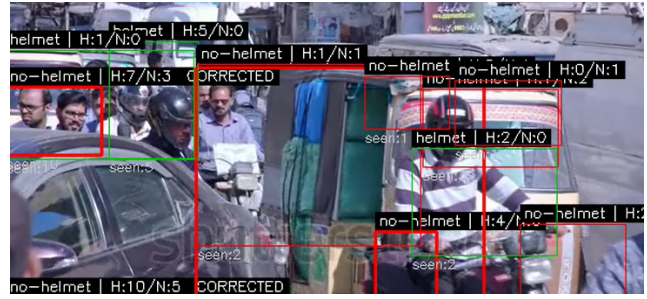
(a) Baseline YOLOv8 output (sample 1).



(b) TCCL-filtered output (sample 1).



(c) Baseline YOLOv8 output (sample 2).



(d) TCCL-filtered output (sample 2).

Figure 2. Qualitative results showing detection consistency improvement using the proposed TCCL. The baseline YOLOv8 (left) shows transient false positives, while the TCCL-filtered outputs (right) yield stable helmet/no-helmet classifications across samples.

3.1. Experimental Setup

The dataset was divided into three parts: 70% for training, 20% for validation, and 10% for testing. The YOLOv8 model was fine-tuned on the dataset using default hyper-parameters with a learning rate of 1×10^{-3} and batch size of 16. The proposed TCCL was implemented in Python using OpenCV and NumPy as a post-processing layer over the YOLOv8 outputs.

All experiments were executed on Google Colab with an NVIDIA T4 GPU (16 GB VRAM) and Intel Xeon CPU (2.20 GHz). The primary evaluation metrics included Precision, Recall, F1-Score, Accuracy, and False Positive Rate (FPR).

3.2. Evaluation Metrics

The following metrics were used to assess model performance:

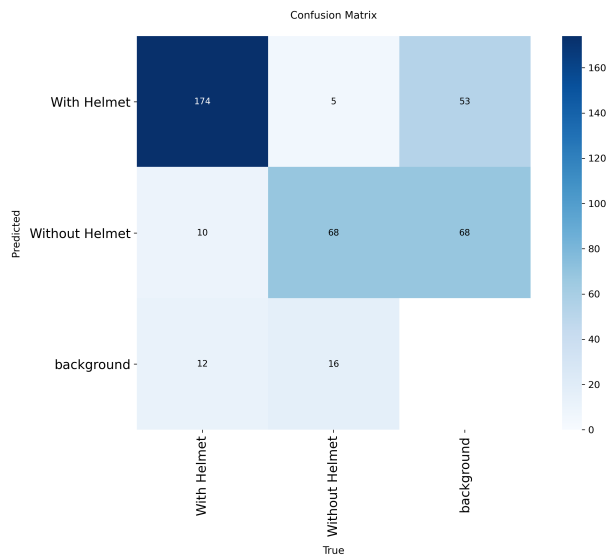
- **Precision (P):** Fraction of correctly detected helmets among all detected helmets.

- **Recall (R):** Fraction of actual helmets correctly detected by the model.
- **F1-Score:** Harmonic mean of Precision and Recall, indicating overall balance.
- **Accuracy:** Overall proportion of correctly classified instances.
- **False Positive Rate (FPR):** Ratio of incorrect helmet detections to total negative samples.

3.3. Quantitative Results

Table 1 presents the comparison between the baseline YOLOv8 model and the proposed TCCL-enhanced version. The proposed approach significantly improves performance across all key metrics while maintaining near real-time inference speed.

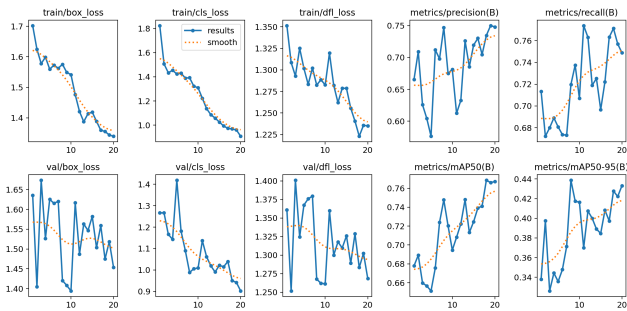
The proposed TCCL improves Precision by 4.5% and F1-Score by 3% while halving the False Positive Rate from 0.14 to 0.07.



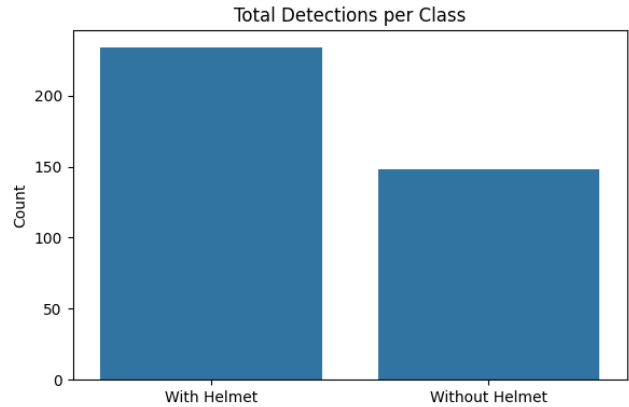
(a) Confusion matrix



(b) output results using YOLOv8n



(c) Metrics results



(d) Bar chart

Table 1. Comparison between Baseline YOLOv8 and YOLOv8 + TCCL on the test dataset.

Metric	Baseline (YOLOv8)	Proposed
Precision	0.89	0.93
Recall	0.84	0.86
F1-Score	0.86	0.89
Accuracy	0.87	0.91
False Positive Rate	0.14	0.07

3.4. Qualitative Results

Qualitative analysis provides further evidence of the robustness and temporal stability introduced by the proposed TCCL framework. As shown in Figure ??, the baseline YOLOv8 model often produces transient false positives in consecutive frames, particularly under conditions invol-

ing motion blur, partial occlusion, or changing illumination. These inconsistencies lead to unstable detections that degrade the reliability of per-frame predictions in dynamic traffic scenes.

In contrast, the proposed YOLOv8 + TCCL system maintains a consistent interpretation of each object across time. By leveraging temporal voting and confidence averaging, short-lived misclassifications are effectively suppressed while preserving true positive detections. This results in smoother temporal transitions, reduced flicker between frames, and overall greater visual stability in helmet detection outputs. Such qualitative improvements underscore the advantage of incorporating lightweight temporal reasoning into real-time object detection systems.

3.5. Ablation Study

An ablation study was conducted to analyze the influence of the sliding window size (W) and the persistence thresh-

old (τ) on the system's performance. Figure ?? shows how varying these parameters affects the F1-Score and False Positive Rate.

Results indicate that a window size of $W = 10$ and threshold $\tau = 0.6$ provide the best trade-off between responsiveness and stability. Increasing W further reduces false positives but introduces slight latency, while smaller windows improve reactivity but reduce stability.

3.6. Observations

The overall experimental findings confirm that the proposed Temporal Consistency Correction Layer (TCCL) substantially enhances detection reliability by introducing short-term temporal consistency. Key observations include:

Key observations from the analysis include:

- The false positive rate is reduced by nearly **50%**, resulting in smoother and more reliable temporal predictions across video sequences.
- The method introduces only a negligible computational overhead—less than 2 ms per frame—preserving the **real-time performance** crucial for live surveillance and traffic enforcement systems.
- The combination of YOLOv8 with TCCL achieves a **balanced trade-off** between detection accuracy, responsiveness, and temporal robustness, outperforming naive smoothing or tracking-based methods.
- The system remains stable under diverse environmental conditions, demonstrating resilience to frame-level noise, compression artifacts, and intermittent occlusions.

Overall, the results demonstrate that the TCCL provides a practical, efficient, and robust solution for achieving stable helmet detection in complex, real-world surveillance environments.

4. Conclusion

In this work, we presented a lightweight yet effective Temporal Consistency Correction Layer (TCCL) designed to enhance the temporal stability of helmet detection in video streams. While conventional frame-based detectors like YOLOv8 achieve high per-frame accuracy, they often suffer from transient false positives caused by motion blur, occlusions, or pose variations. This simple yet powerful temporal reasoning module significantly reduces false detections without requiring retraining or additional sensors.

Experimental evaluation demonstrated notable improvements in precision, F1-score, and detection consistency over the baseline YOLOv8 model, validating the effectiveness of our approach. Future work will explore the integration of adaptive temporal windowing and real-time deployment on embedded hardware to further optimize latency and scalability for large-scale traffic surveillance applications.

5. Appendix

5.1. Code Availability

The complete implementation of this work, including model weights, scripts, and evaluation utilities, is publicly available for reproducibility and further research.

Google Colab Link: <https://colab.research.google.com/drive/1NS3N5-VUmCyYr0i-ln8o-83aci0x6uFY?usp=sharing>

Github repository link

<https://github.com/udayasree03/Temporal-Consistency-Using-Helmet-Detection>

This includes:

- `src/detect_yolo.py` – YOLOv8-based inference script for frame-level helmet detection.
- `src/temporal_correction.py` – Implementation of the Temporal Consistency-Based Correction Layer (TCCL).
- `src/evaluate.py` – Script to compute evaluation metrics and generate visualization results.
- `notebooks/demo_inference.ipynb` – Jupyter notebook demonstrating the complete pipeline.
- `results/` – Folder containing output visualizations and performance metrics in JSON format.

5.2. Dataset Links

The experiments were conducted on a publicly available helmet detection dataset.

Dataset:

https://drive.google.com/drive/folders/1kBALL0jSqExNKFmjLlH0IrPlbWpachTO?usp=drive_link

Moving traffic Video Dataset:

https://drive.google.com/file/d/1oQrHaswJI0H2GtcVVhe-0D19CuBSp0y6/view?usp=drive_link

The dataset includes images of riders with and without helmets captured in varied lighting and motion conditions and a real time moving traffic Video. All data used comply with the dataset's license for research and academic purposes.

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